

Number of Players Zero

AI Tic Tac Toe via LLMs

V2 05-24-26
Revis



Stephen Witty

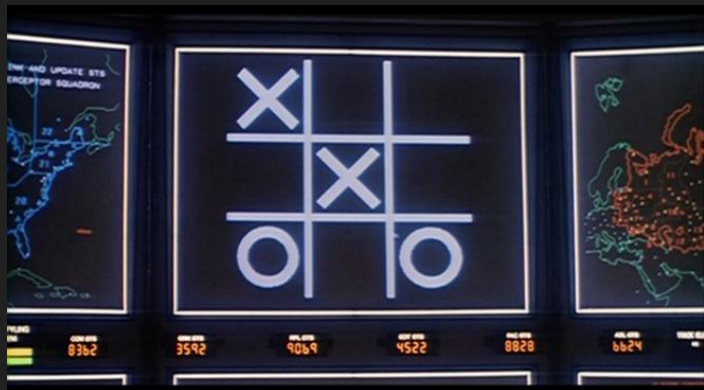
switty@level500.com

GitHub: level500.com

870-688-5220

Movie WarGames - 1983

Project inspiration



Project Goal

Test LLMs playing Tic Tac Toe

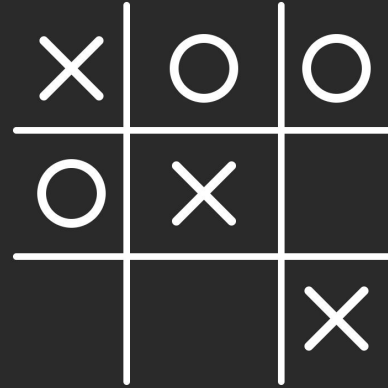
Measure results - wins, ties, draws

Multiple autonomous game plays

LLM to same LLM

LLM to perfect opponent

LLM to different LLM



LLM prompt

You are playing Tic-Tac-Toe. Your symbol is X.

Board positions:

1 | 2 | 3

---+---+---

4 | 5 | 6

---+---+---

7 | 8 | 9

Current board (X=you, O=opponent, number=empty):

O | 2 | O

---+---+---

4 | X | 6

---+---+---

7 | 8 | X

Available squares: 2, 4, 6, 7, 8

Reply with ONLY your move in this exact format: {N}

No explanation, no commentary, no other text. Just {N}.

LLM Tic-Tac-Toe Battle

OPENROUTER CONNECTION

API Key

.....

Load Available Models

✓ 358 models loaded (28 free)

TOURNAMENT SETTINGS

Number of Games

5

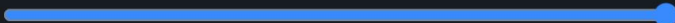
Max API Errors

10

Max Format Errors

10

Temperature **1.00**



PLAYERS

Player X — Model

☐ Perfect Player (Algorithm)

☐ Free models only **FREE**

Search models...

Anthropic: Claude Haiku 4.5 (anthropic/claude-haiku-4.5) ▾

Player O — Model

☐ Perfect Player (Algorithm)

☐ Free models only **FREE**

Search models...

OpenAI: GPT-5.4 Mini (openai/gpt-5.4-mini) ▾

Start Tournament

LLM Tic-Tac-Toe Battle

5/5

GAME

4

X WINS

1

O WINS

0

DRAWS

0

API ERRORS

0

X FMT ERRORS

0

O FMT ERRORS

\$0.003477

X COST

\$0.002151

O COST

5

UNIQUE GAMES

Board

Tournament complete

X: anthropic/claude-haiku-4.5

O: openai/gpt-5.4-mini

O	2	X
X	X	X
7	O	O

X's Turn

O's Turn

New Tournament

Save Results

Game Log

Board positions:

```
1 | 2 | 3
---+---+---
4 | 5 | 6
---+---+---
7 | 8 | 9
```

Current board (X=you, O=opponent, number=empty):

```
0 | 2 | X
---+---+---
X | X | 6
---+---+---
7 | 0 | 0
```

Available squares: 2, 6, 7

Reply with ONLY your move in this exact format: {N}

No explanation, no commentary, no other text. Just {N}.

20:32:49 X (anthropic/claude-haiku-4.5) responded: "{6}"

20:32:49 X plays square 6

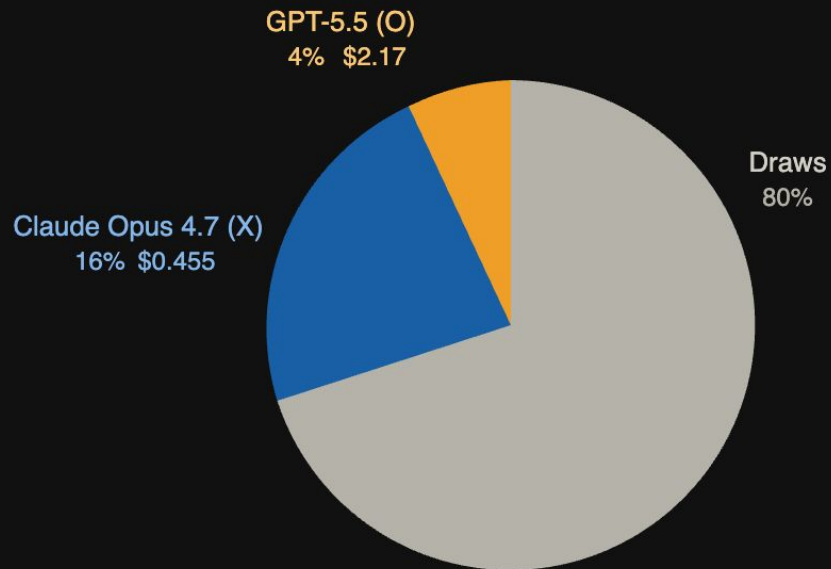
20:32:49 X wins Game 5!

20:32:51 — Tournament complete —

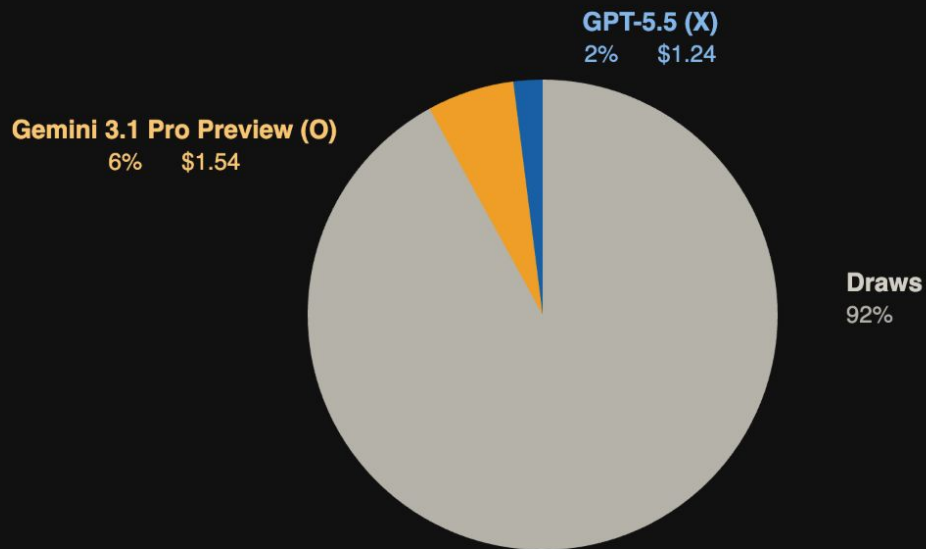
20:32:51 X wins: 4 | O wins: 1 | Draws: 0

Demo Video

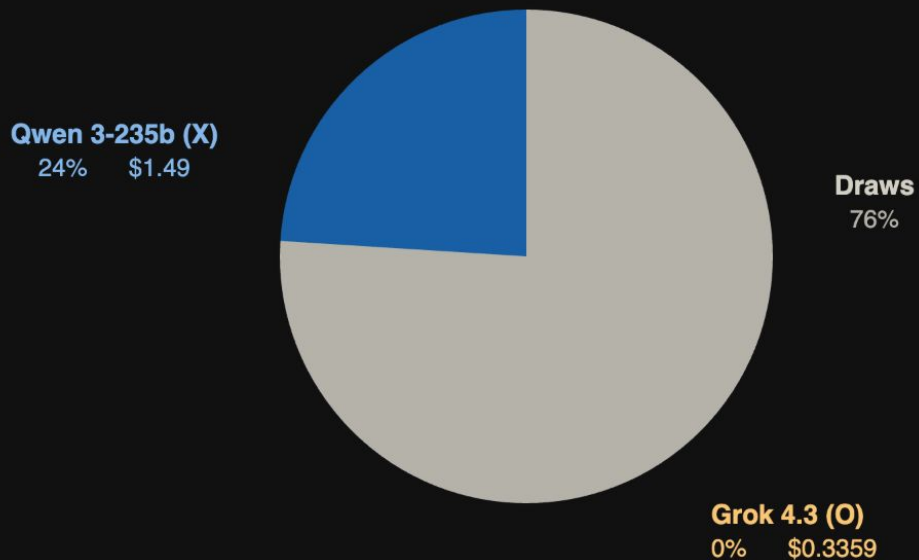
Opus 4.7 vers GPT 5.5 (Wins)



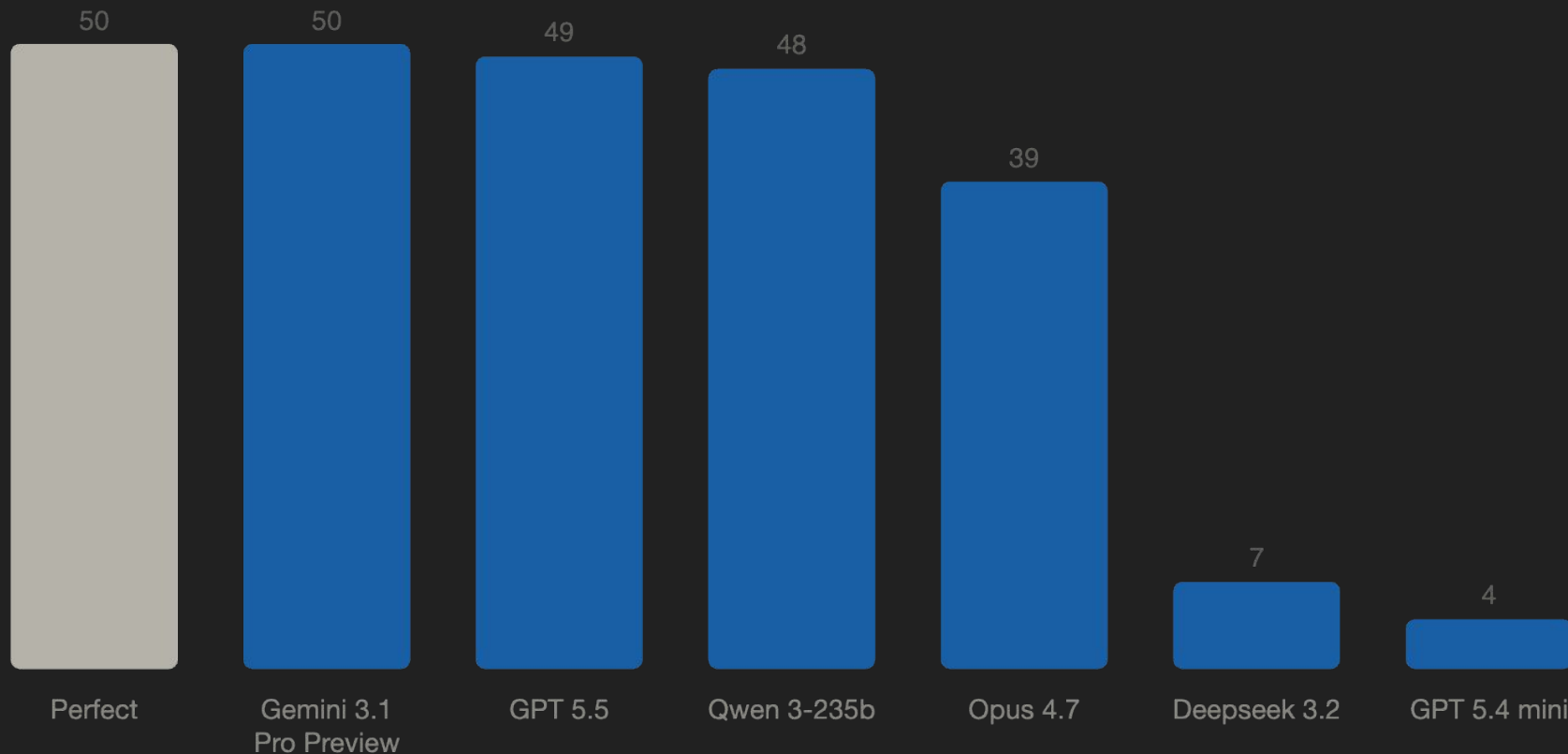
Gemini 3.1 Pro Preview vers GPT 5.5 (Wins)



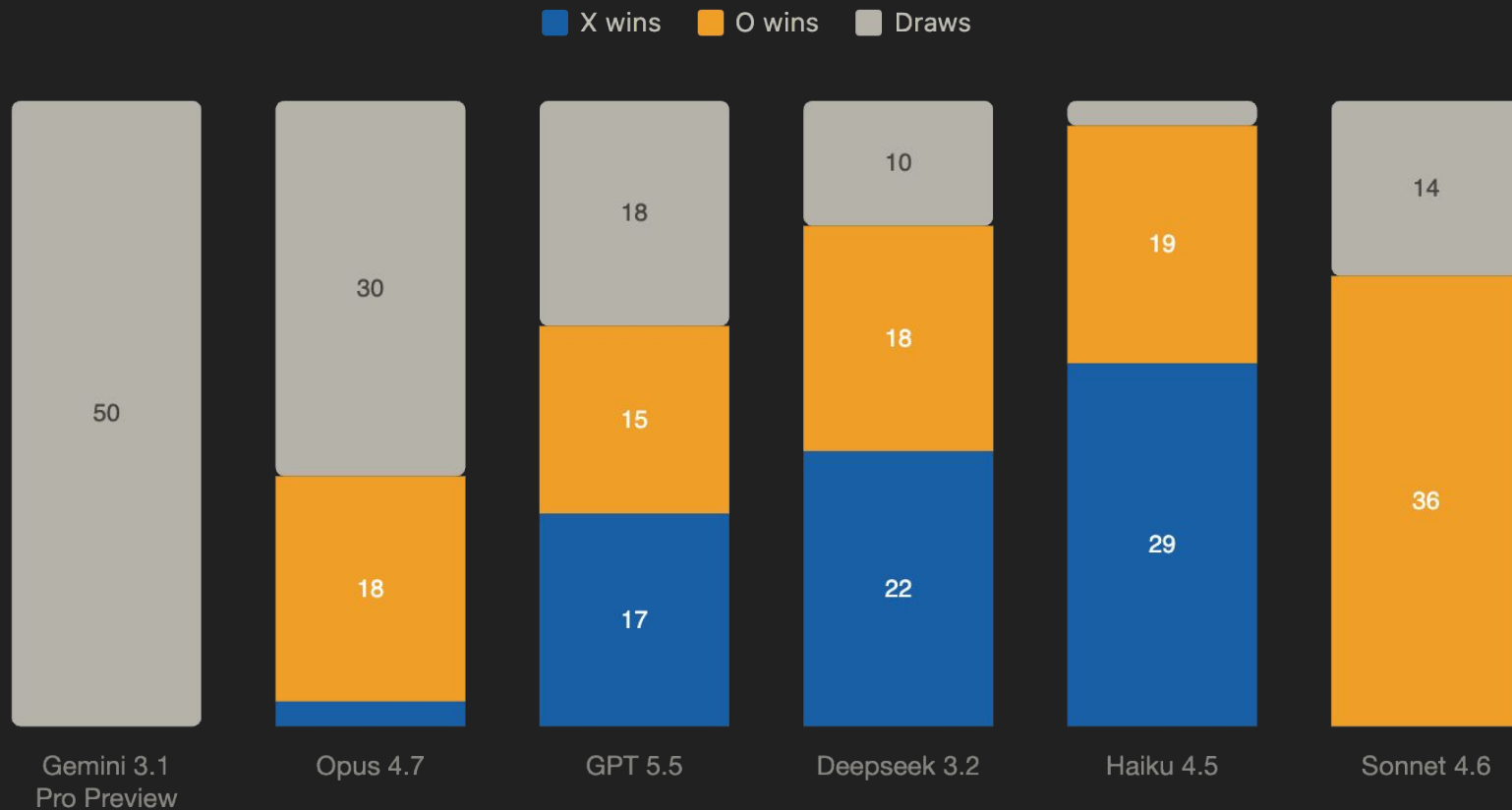
Qwen 3-235b vers Grok 4.3 (Wins)



Number of draws for LLM against perfect player



LLM against same LLM



Conclusions

Not all LLMs play well

X and O are not equal in game play

Open Router free LLMs - heavy throttling

Hard to produce unique games of Tic Tac Toe

Questions?



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